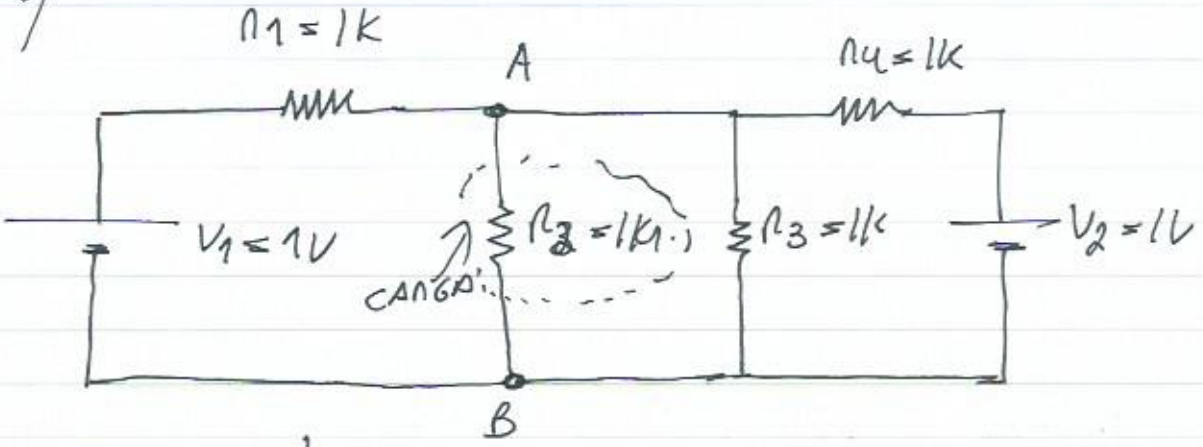
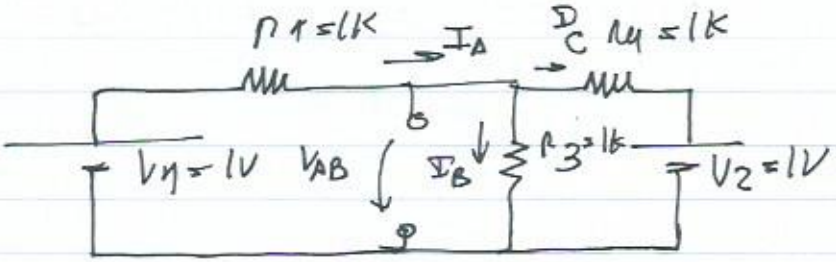
13/ (8)

a/



Calculo Vth = ?



$$\begin{cases} -V_1 + R_1 I_A + R_3 I_B = 0 \\ R_4 I_C + V_2 - R_3 I_B = 0 \\ I_A = I_B + I_C \end{cases} \Rightarrow \begin{cases} R_1 I_A + R_3 I_B + 0 = V_1 \\ 0 - R_3 I_B + R_4 I_C = -V_2 \\ I_A - I_B - I_C = 0 \end{cases}$$

$V_{th} = V_{AB} = R_3 \times I_B$

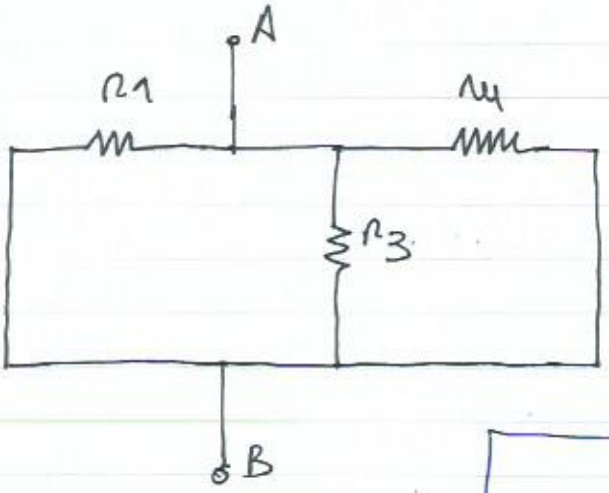
$$\begin{bmatrix} R_1 & R_3 & 0 \\ 0 & -R_3 & R_4 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \\ 0 \end{bmatrix}$$

$$V_{th} = R_3 \times I_B = \underline{9{,}6\,V}$$

AULA 5

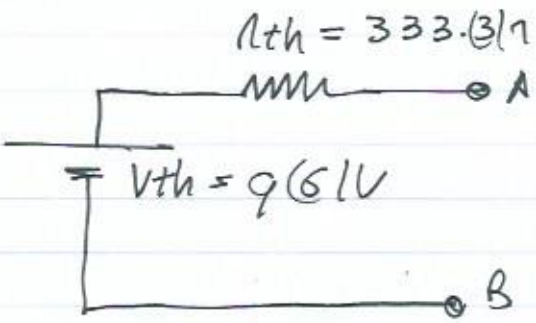
(9)

Calcule R_{th} = ?



$R_{th} = R_1 // R_3 // R_4$

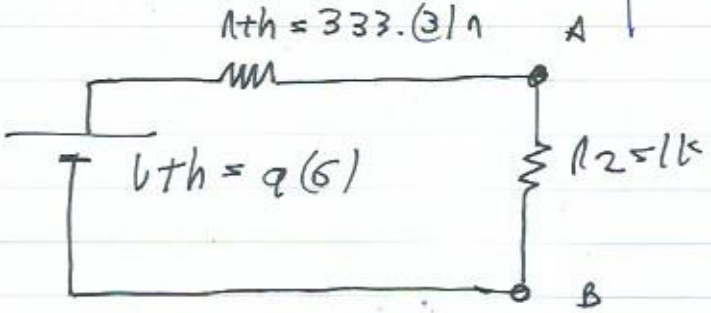
$R_{th} = 1k // 1k // 1k = 93k$



b) SIMULAÇÃO:

c)

SITUAÇÃO ORIGINAL:



$-V_{th} + R_{th} I_{NOVO} + R_{2\,NOVO} I_{NOVO} = 0$

$R_{2\,NOVO} I_{NOVO} = V_{th} - R_{th} I_{NOVO}$

$R_{2\,NOVO} = \dfrac{V_{th} - R_{th} I_{NOVO}}{I_{NOVO}}$

$R_{2\,NOVO} = \dfrac{9(6) - 333.(3) \times 9{,}25m}{9{,}25m}$

$I_{R_2\,INICIAL} = \dfrac{9(6)}{333.(3) + 1k.}$

$I_{R_2\,INICIAL} = 0{,}0005\,A = 9{,}5mA$

$I_{R_2\,FINAL} = \dfrac{1}{2} I_{R_2\,INICIAL} = 9{,}25mA$

$$I_{R_2\ \text{FINAL}} = 25\,mA = \frac{V_{th}}{R_{2\ \text{NOVO}} + R_{th}} = \frac{6\,(1)}{R_{2\ \text{NOVA}} + 2\,(3)\,k}$$

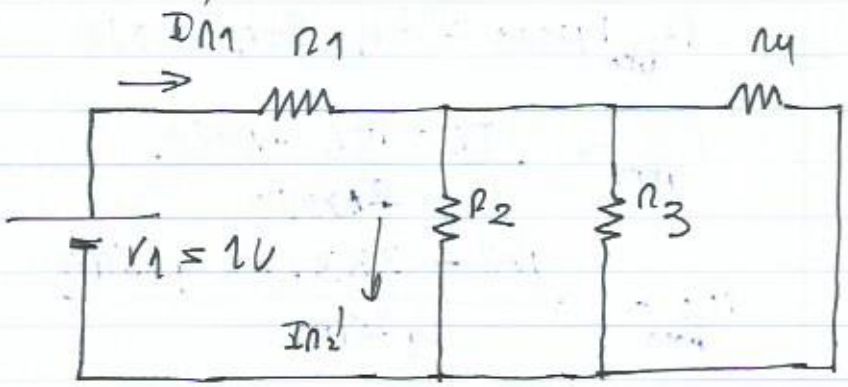
$$25\,mA \times \left(R_{2\ \text{NOVA}} + 333.(3)\right) = 6\,(1)$$

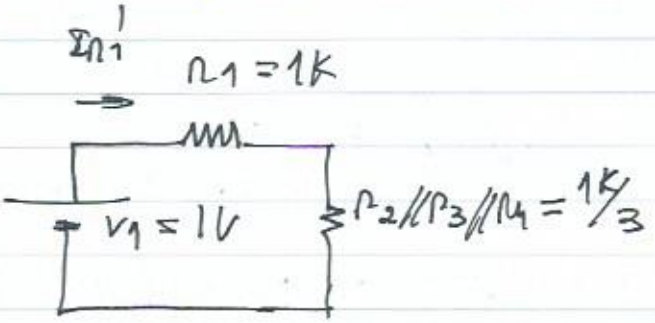
$$R_{2\ \text{NOVA}} + 333.(3) = \frac{6\,(1)}{25\,mA} \Rightarrow R_{2\ \text{NOVA}} = \frac{6\,(1)}{25\,mA} - 333.(3)$$

$$R_{2\ \text{NOVA}} = 2333.(3)\,\Omega = 2.(3)\,k\Omega$$

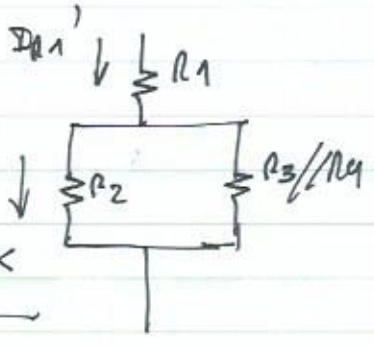
d) TEOREMA DA SOBREPOSIÇÃO.

CONTRIBUTO V_1





$$I_{R_1}' = \frac{1}{1k + 1k/3} = \frac{1}{1.(3)\,k}$$



$$I_{R_2}' = \frac{I_{R_1}' \times R_3 // R_4}{R_3 // R_4 + R_2} = \frac{\frac{1}{1.(3)\,k} \times 9.5\,k}{9.5\,k + 1k}$$

$$= 25\,mA$$

Nome _____ N.º Aluno _____

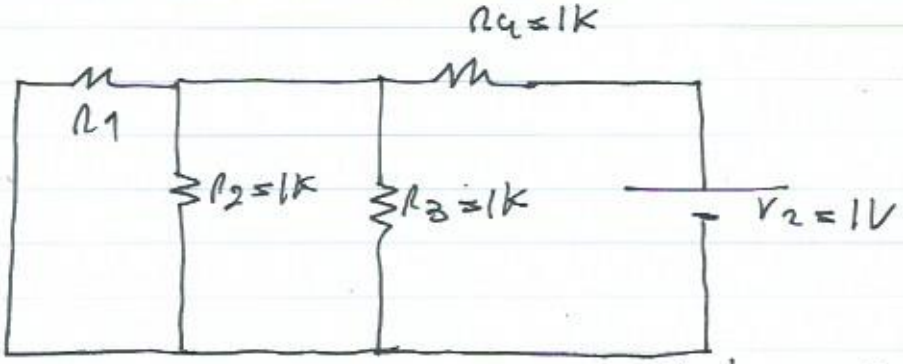
Curso _____ Ano Letivo _____ / _____ Data da Avaliação _____ / _____ / _____

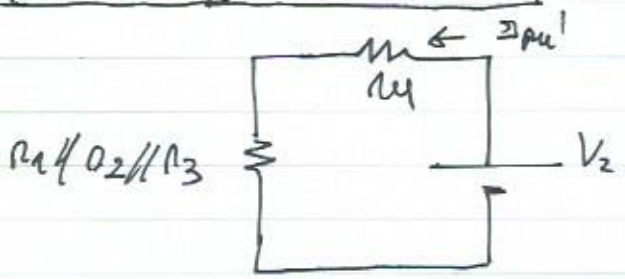
Prova Escrita de: _____ N.º Folhas _____

Época: _____

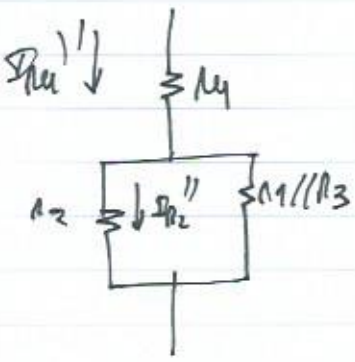
(11)

CONTRIBUTO V_2 (IGUAL) => CIRCUITO SIMÓTICO





$$I_{R4}'' = \frac{V_2}{R_4 + R_1 // R_2 // R_3}$$

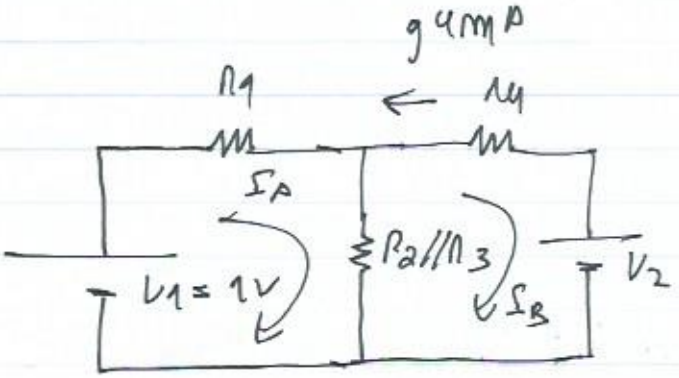


$$I_{R2}'' = \frac{I_{R4}'' \times (R_1 // R_3)}{R_1 // R_3 + R_2} = 9{,}25\,mA$$

$$I_{R2} = I_{R2}' + I_{R2}'' = 9{,}5\,mA$$

c) $P_{R4} = R_4 \times I_{R4}^2 = ?$

Calculo $I_{R4} = ?$



$$\begin{cases} -V_1 + R_1 I_A + R_2 /\!/ R_3 (I_A - I_B) = 0 \\ R_4 I_B + V_2 + R_2 /\!/ R_3 (I_B - I_A) = 0 \end{cases}$$

$$I_A (R_1 + R_2 /\!/ R_3) + I_B (-R_2 /\!/ R_3) = V_1$$

$$I_A (-R_2 /\!/ R_3) + I_B (R_4 + R_2 /\!/ R_3) = -V_2$$

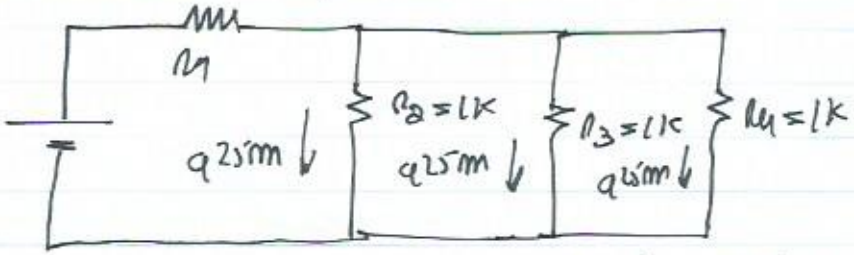
$$\begin{bmatrix} (1k + 0.5k) & (-0.5k) \\ (-0.5k) & 1k + 0.5k \end{bmatrix} \begin{bmatrix} I_A \\ I_B \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$I_B = -0.5\,mA$$

TEOREMA SOBREPOSICIÓN

CONTRIBUCIÓN V_1

$I'_{R4} = 0.75\,mA$



$I'_{R4} = 0.25\,mA$ (→)

CONTRIBUCIÓN I_1

I_{R4} (←)

$I''_{R4} = \frac{1}{1.3k} = 0.75\,mA$

$I_{R4} = -I'_{R4} + I''_{R4} = 0.5\,mA$ (←)

$P_{14} = 1k \cdot (0,5)^2 = 925\ mJ$

$E_J = 925\ mJ \times 30 \times 24 \times 3600 = 648\ J.$

f) $t = \frac{E}{P} = \frac{1000\ J}{925\ m} = \frac{1000}{925 \times 10^{-3}} = 4 \times 10^6\ s$

1 dia — $24 \times 60 \times 60$ s

x dias — 4×10^6 s

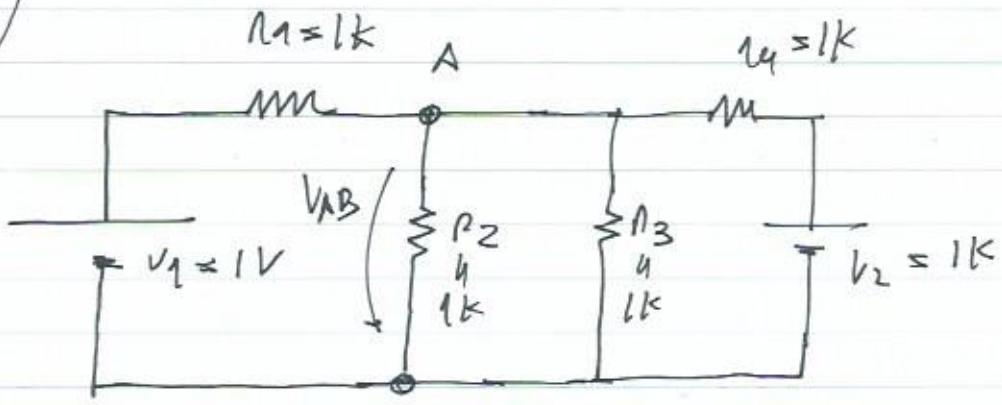
$x_{DIAS} = \frac{4 \times 10^6}{24 \times 3600} \simeq 46.3$ dias.

$\frac{365}{12}$ dias — 1 mês $\Rightarrow x_{MESES} = \frac{46.3}{365} \times 12$

46.3 dias — 2 meses

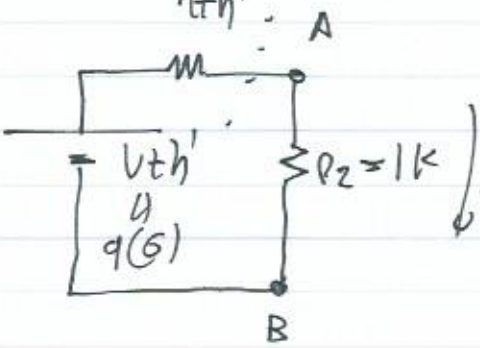
$x_{MESES} \simeq 1.52$

MÊS e MEIO



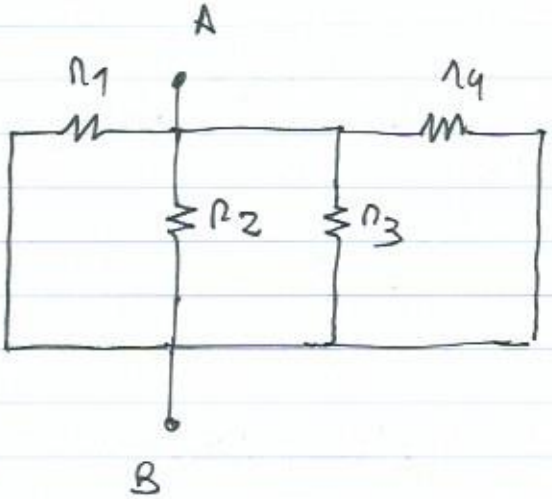
333.(3)
"
Rth'

Vth' e Rth' → ALUNGA (A)



$$V_{AB} = \frac{R_2 \times V_{th}'}{R_{th}' + R_2}$$

$$V_{AB} = \frac{1k \times 9(6)}{333.(3) + 1k} = 9{,}5V$$



$$R_{th} = R_1 // R_2 // R_3 // R_4 = 250\Omega$$

